Voronoi tessellations for protein interfaces and SEPIQ

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Where I am from

https://gmc.vu.lt

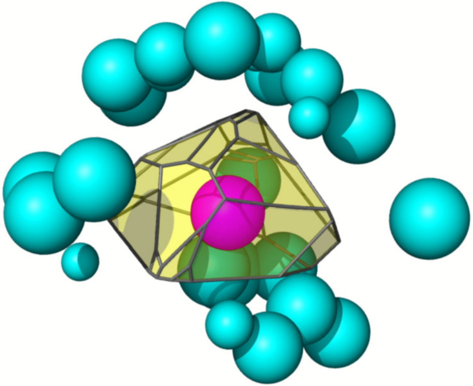
https://bioinformatics.lt

What I do

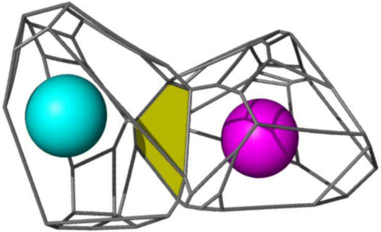
- I am a senior researcher in Vilnius University Life Sciences Center, I work on developing novel effective methods of **structural bioinformatics**.
- Most of my methods construct and utilize the **Voronoi tessellation** of atomic balls and the tessellation-derived interatomic contact areas.
- The developed tessellation-based methodology is especially efficient in analyzing **protein-protein interfaces**.

Deriving atom-atom contact areas using the Voronoi tessellation

Voronoi cell of an atom
surrounded by its neighbors

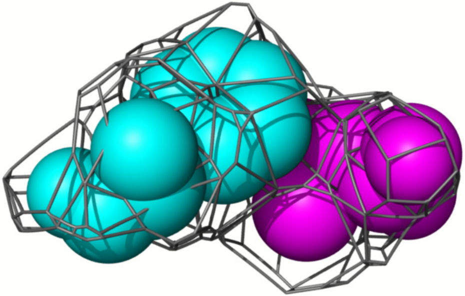


Atom-atom contact surface
defined as the face shared
by two adjacent Voronoi cells.

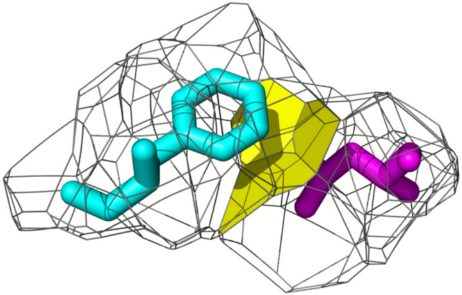


Deriving residue-residue contacts

Voronoi cells of two neighboring residues

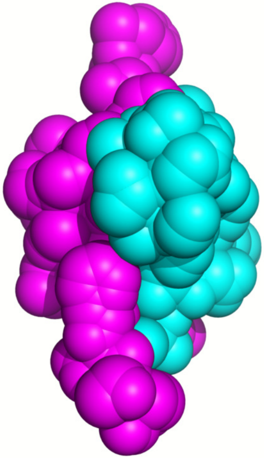


Residue-residue contact surface defined as a union of atom-atom contact surfaces

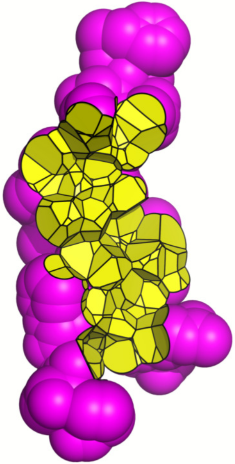


Inter-chain contacts

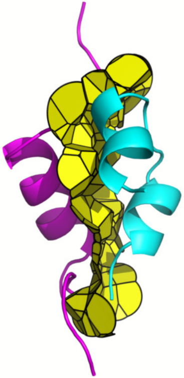
Solvent-accessible surface of an insulin heterodimer PDB:4UNG colored by subunit



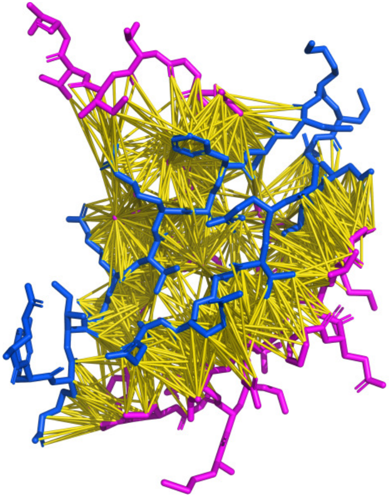
The intersubunit interface shown together with the SAS of one subunit



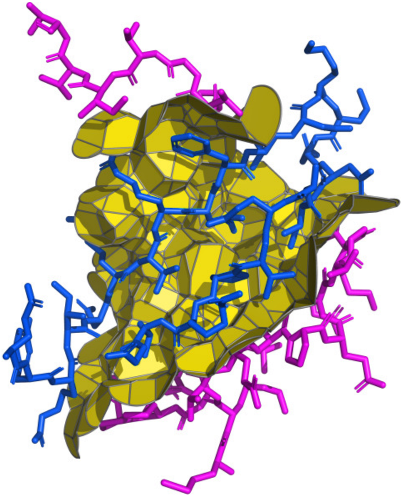
The intersubunit interface shown together with both subunits represented as cartoons



Inter-chain contact areas vs distances

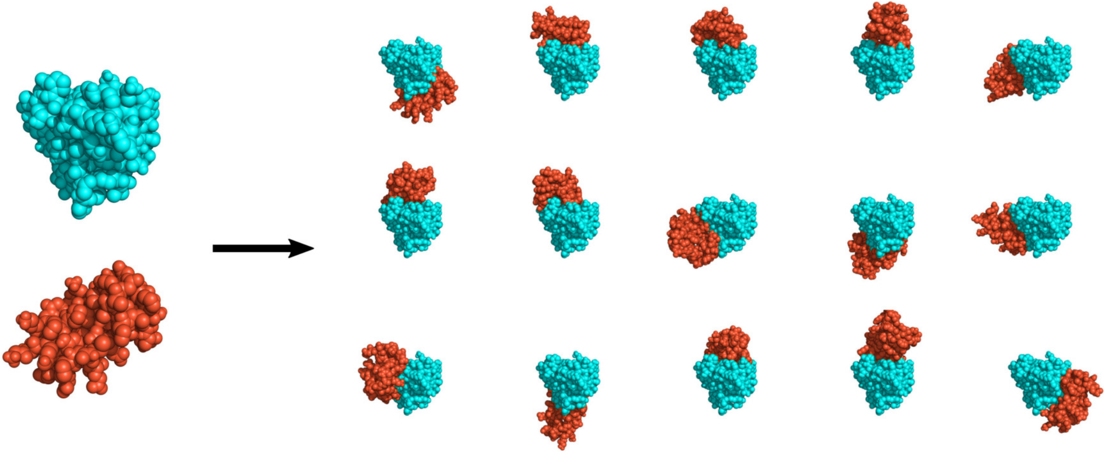


VS

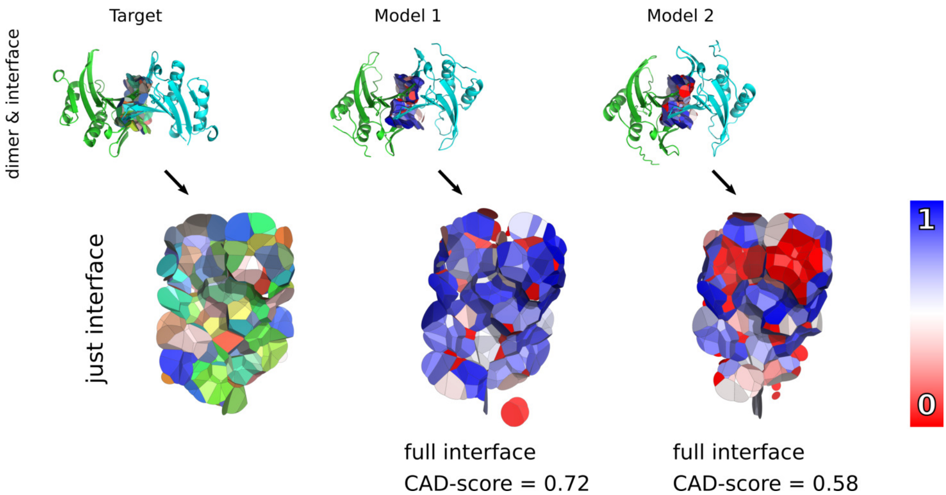


Olechnovic and Grudinin. *Voronota-LT: Efficient, Flexible, and Solvent-Aware Tessellation-Based Analysis of Atomic Interactions*. JCC (2025)

Same chains can have differently modelled interfaces

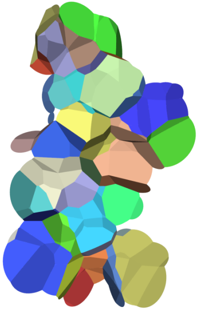


Comparing interfaces using CAD-score (Contact Area Difference score)



Olechnovic and Venclovas. *Contact Area-Based Structural Analysis of Proteins and Their Complexes Using CAD-Score*. Methods in Molecular Biology (2020)

Evaluating interfaces with an area-based potential (e.g. VoroMQA)



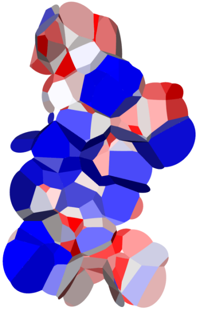
Interface contact areas

$$\mathrm{E}(a_i, a_j, c_k) = \log \frac{\mathrm{P}_{\mathrm{exp}}(a_i, a_j, c_k)}{\mathrm{P}_{\mathrm{obs}}(a_i, a_j, c_k)} =$$

$$= \log \frac{\mathrm{F}_{\mathrm{exp}}(\mathrm{area}(a_i), \mathrm{area}(a_j), \mathrm{area}(c_k))}{\mathrm{F}_{\mathrm{obs}}(\mathrm{area}(a_i, a_j, c_k))}$$

$$\mathrm{E_n}(\Omega_\phi) = \frac{\sum_{\omega \in \Omega_\phi} \mathrm{E}(\mathrm{type}_\omega) \cdot \mathrm{area}_\omega}{\sum_{\omega \in \Omega_\phi} \mathrm{area}_\omega}$$

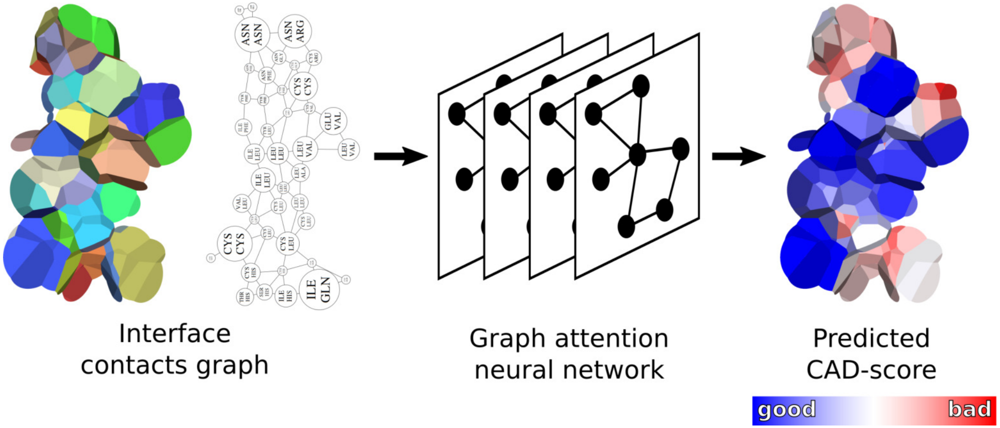
Statistical potential for contact areas



Interface pseudo-energy

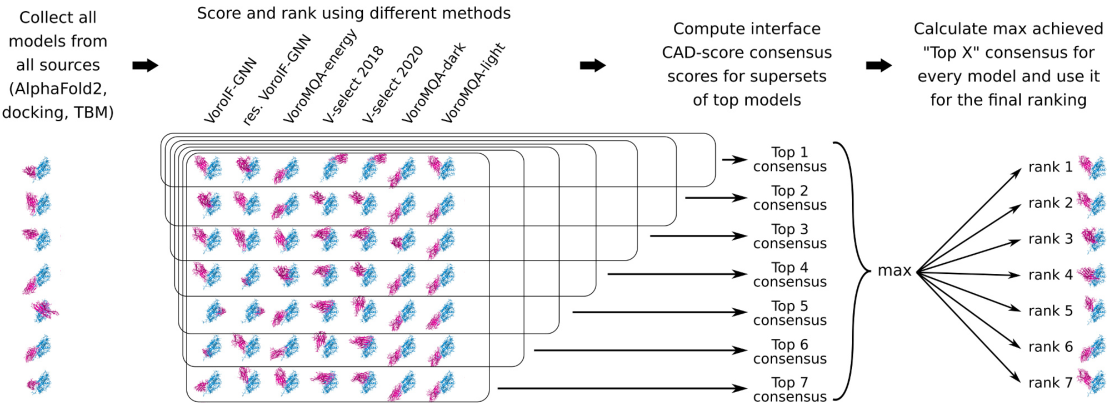
Olechnovic and Venclovas. *VoroMQA: Assessment of protein structure quality using interatomic contact areas*. Proteins (2017)

Evaluating interfaces with a graph neural network (e.g. VoroIF-GNN)



Olechnovic and Venclovas. *VoroIF-GNN: Voronoi tessellation-derived protein-protein interface assessment using a graph neural network*. Proteins (2023)

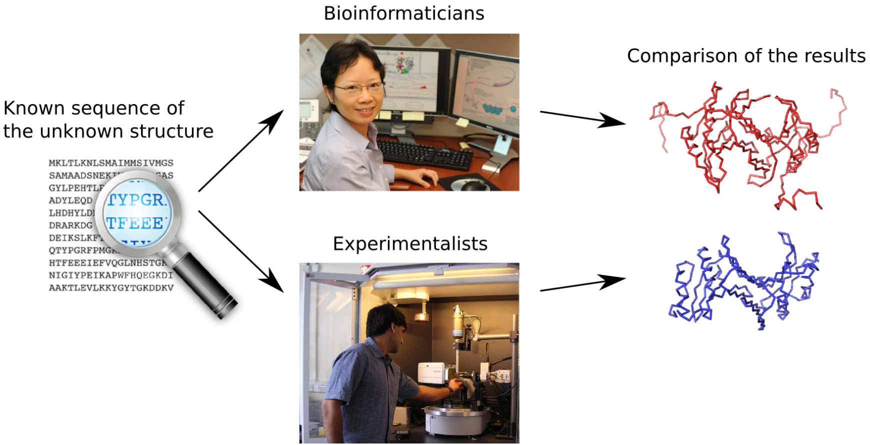
Selecting complex models using VoroIF-jury (a.k.a. FTDMP)



VoroIF-jury was the best-performing multimeric model selector in 2022 CASP and CAPRI double-blind challenges.

CASP and CAPRI double-blind challenges

The main way to test new methods in structural bioinformatics are the community-wide double-blind testing experiments — CASP and CAPRI.



Our results in CASP and CAPRI challenges over the years

CASP (Critical Assessment of Techniques for Protein Structure Prediction) and CAPRI (Critical Assessment of PRedicted Interactions) are world-wide experiments focused on the blind testing of methods for protein structural bioinformatics.

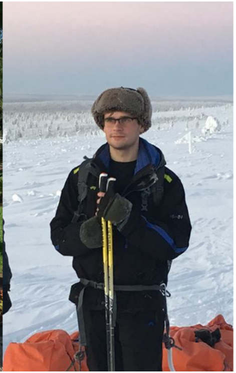
After AF2

2024	Top performance (ranked 1st in the CAPRI assessment of scoring groups) in the CASP16-CAPRI scoring experiment. Group "Olechnovic".
2024	Contributed to the top performance (ranked 1st) in modeling structures of protein complexes in the CAPRI experiment rounds 47–55. Group "Venclovas", members: Dapkūnas J, Olechnovič K, Venclovas Č.
2022	Contributed to one of the top performances in modeling structures of protein complexes in CASP15 and CASP15-CAPRI experiments (ranked 2nd in CASP, ranked 1st in CAPRI). Group "Venclovas", members: Olechnovič K, Valančauskas L, Dapkūnas J, Venclovas Č.
2022	One of the top performances in EMA (estimation of model accuracy) in CASP15 experiment (and ranked 1st in CASP15-CAPRI scoring experiment). Groups "Venclovas" and "VoroIF".

Before AF2

2020	Contributed to one of the top performances in modeling structures of protein complexes in CASP14 and CASP14-CAPRI experiments (ranked 2nd in CASP, ranked 1st in CAPRI jointly with two other groups). Group "Venclovas", members: Olechnovič K, Dapkūnas J, Venclovas Č.
2019	Contributed to one of the top performances (ranked 3rd) in modeling structures of protein complexes in the CAPRI experiment rounds 38–45. Group "Venclovas", members: Dapkūnas J, Kairys V, Olechnovič K, Venclovas Č.
2018	Contributed to the best results (ranked 1st) in modeling structures of protein complexes in CASP13 and CASP13-CAPRI experiments. Group "Venclovas", members: Dapkūnas J, Olechnovič K, Venclovas Č.
2018	One of the top performances in EMA (estimation of model accuracy) in CASP13 experiment (ranked 1st in prediction of unreliable regions). Groups "VoroMQA-A" and "VoroMQA-B".
2016	Contributed to the best results (ranked 1st) in modeling structures of protein complexes in CASP12-CAPRI experiment. Group "Venclovas", members: Dapkūnas J, Olechnovič K, Venclovas Č.
2016	One of the top performances in protein structure prediction in CASP12 experiment. Group "VoroMQA-select".

CASP and CAPRI team



CASP15 Team

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Life Sciences Center

Institute of Biotechnology

Funding: Research Council of Lithuania

What I do with SEPIQ

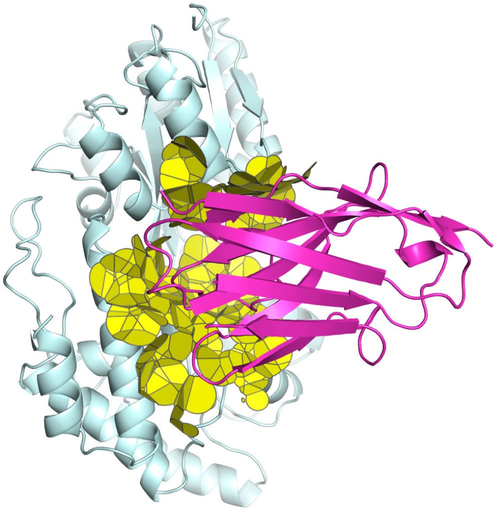
I am involved in the development of the framework for assessing predictions of antibody-antigen interface residues.

More, specifically, I consult on:

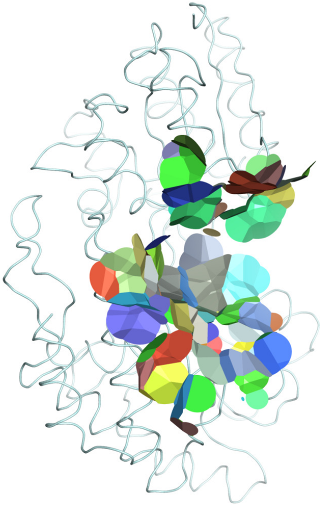
- how to describe the interface binding sites in the ground truth structure
- how to evaluate the prediction of the binding sites based on the ground truth description

SEPIQ-like example of structural interface ground truth

Inter-chain contact areas
in ground truth structure



The same contact areas
grouped per antigen residue



SEPIQ predictions assessment outline

1. The ground truth 3D structures will be converted to binary per-residue labels: binding or non-binding (1 or 0).
2. This makes the prediction task for SEPIQ Track A a binary classification task: assign labels to sequence positions to match ground truth labels as well as possible.
3. The ground truth data will likely be highly imbalanced - there will be many more non-interface residues than interface residues.
4. The methodologies for evaluating binary classification results for imbalanced data are well known, one of the most straightforward ways is to use PR AUC.
5. For an additional evaluation, it is possible to use ground truth per-residue binding site area in the assessment to put more emphasis on residues that are more involved in the interface.

Thank you

Thank you SEPIQ team.

Thank you all for listening.